

Lexical Semantics

Week 7

Slides and materials based on the courses by
Noortje Venhuizen and Mareike Hartmann

A closer look at plural NPs

- Entailment pattern (of predicates): **distributivity**
- “*Bill and Mary work*” $\models$ “*Bill works*”
 - $work(b) \wedge work(m) \models work(b)$
- “*Bill and Mary work*” $\models$ “*Mary works*”
 - $work(b) \wedge work(m) \models work(m)$
- “*all students work*”, “*John is a student*” $\models$ “*John works*”
 - $\forall x(student(x) \rightarrow work(x)), student(j) \models work(j)$

Distributivity does not hold for all predicates

- “*Bill and Mary met*” $\not\models$ “*Bill met*”
- “*the students met*”, “*John is a student*” $\not\models$ “*John met*”
- “*the committee will dissolve*”, “*John is member of the committee*”
 $\not\models$ “*John will dissolve*”
- “*meet*” and “*dissolve*” are **collective** predicates

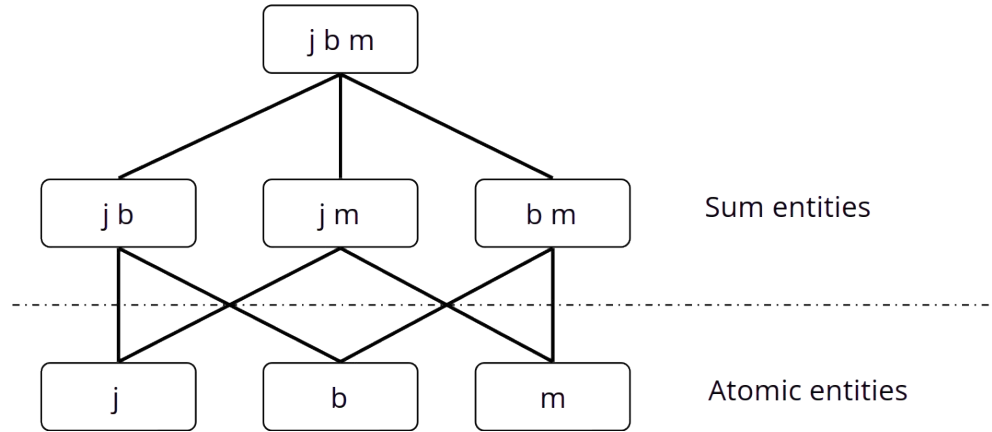
Distributive vs. collective predicates

- Distributive:
 - Applicable to singular and plural NPs
 - Predication with a plural NP *distributes* over the individual objects covered by the NP
 - Examples: “*work*”, “*sleep*”, “*eat*”, “*tall*”, ...
- Collective:
 - Only applicable to plural or group NPs
 - Semantics cannot be reduced to atomic statements about single standard individuals
 - Examples: “*meet*”, “*gather*”, “*unite*”, “*disperse*”, “*dissolve*”, ...
- **Mixed** predicates (“*carry a piano*”, “*solve the exercise*”): predicates that are ambiguous between the distributive and collective reading

Modeling plural terms: desiderata

- A representation of plural terms that is not (only) defined in terms of atomic entities (to account for collective predicates)
 - We extend the universe of our model structures with **sums** (or: “groups”)
- A relation between atomic and plural entities (to account for the entailment pattern of distributive predicates)
 - We add a membership (or: “individual part”) relation to the model structure
- Denotations of types of predicates are restricted to particular parts of universe

Structured universe: entities and sums of entities



- Edges indicate the (individual) **part-of relation**

Algebraic detour: partial orders

- A **partial order** is an algebraic structure $(A, \leq)$ where $\leq$ is a *reflexive*, *transitive*, and *antisymmetric* relation over A .
- The **join** $(a \sqcup b)$ of $a, b \in A$ is the *lowest upper* bound for a and b
 - $a \sqcup b = x$ s.t. $a \leq x$ and $b \leq x$ and $\neg \exists y (y \neq x \wedge (a \leq y \leq x \vee b \leq y \leq x))$
- The **meet** $(a \sqcap b)$ of $a, b \in A$ is the *highest lower* bound for a and b
 - $a \sqcap b = x$ s.t. $x \leq a$ and $x \leq b$ and $\neg \exists y (y \neq x \wedge (x \leq y \leq a \vee x \leq y \leq b))$

Algebraic detour: lattices and semi-lattices

- A lattice is a partial order $(A, \leq)$ that is **closed** under meet and join
 - i.e. for all $a, b \in A$: $(a \sqcup b) \in A$ and $(a \sqcap b) \in A$
 - Examples:
 - Boolean algebra
 - *complemented distributive **bounded*** lattice: $(A, \leq, \perp, \top, \neg)$
 - Powerset **Boolean algebra**: $(\mathcal{P}(S), \subseteq, \emptyset, S, S - (-))$
 - Integers, real numbers
- A **join semi-lattice** is a partial order $(A, \leq)$ that is closed under join

Algebraic detour: bounded lattices

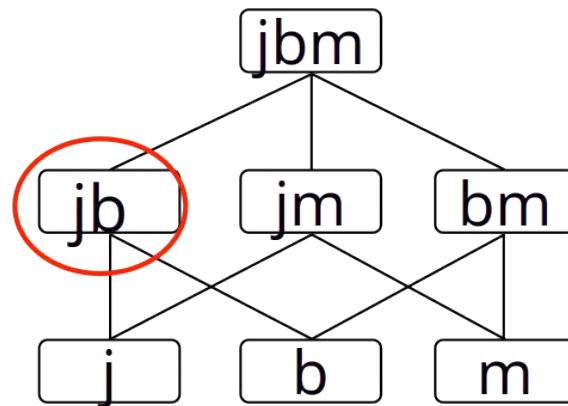
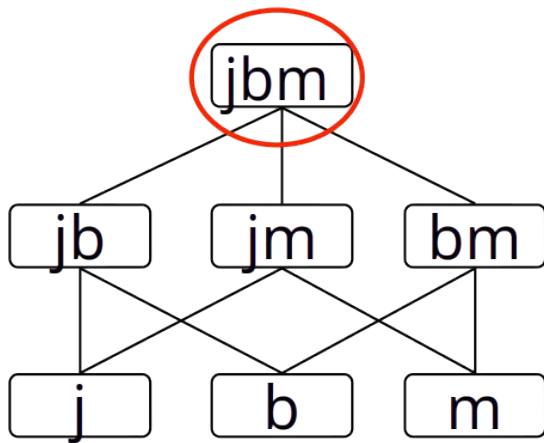
- A **bounded lattice** has a maximal element (1) and a minimal element (0)
- An element $a \in A$ is an **atom**, if $a \neq 0$ and there is no $b \neq 0$ in A such that $b < a$
- A lattice is **atomic**, if for every $b \neq 0$ there is an atom a such that $a \leq b$

Model structures for plural terms

- A model structure is a pair $M = ((U, \leq), V)$, where
 - $(U, \leq)$ is an *atomic join semi-lattice* over the universe U , and $\leq$ is the **individual part relation**
 - V is an interpretation function
- In addition, we define:
 - $A \subseteq U$ is the set of atoms in $(U, \leq)$
 - $U - A$ is the set of non-atomic elements, i.e. the set of **proper sums** (or “groups”) in U

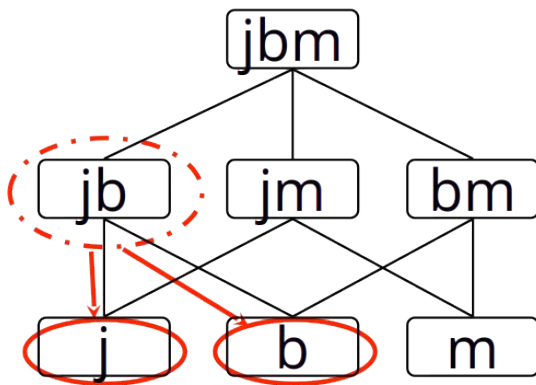
Domain restrictions: collective predicates

- Let P_c be the set of collective predicates (*“meet”*, *“dissolve”*, etc.)
 - The domain of P_c is restricted to non-atomic elements:
for all $R \in P_c$, $V_M(R) \subseteq U - A$

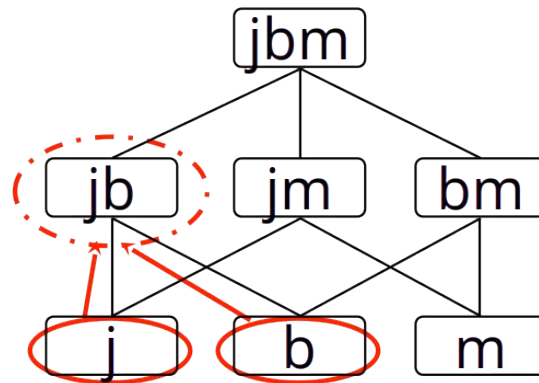


Domain restrictions: distributive predicates

- Let P_d be the set of distributive predicates (“work”, “tall”, “student”, etc.)
 - The domain of P_d is the universe of M : for all $R \in P_d$, $V_M(R) \subseteq U_M$, such that $a \in V_M(R)$ and $b \in V_M(R)$ iff $a \sqcup b \in V_M(R)$



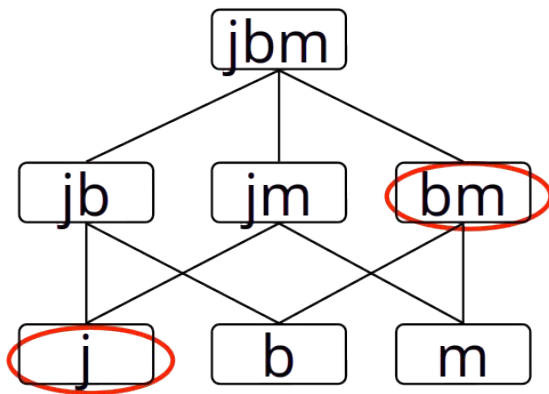
Distributivity



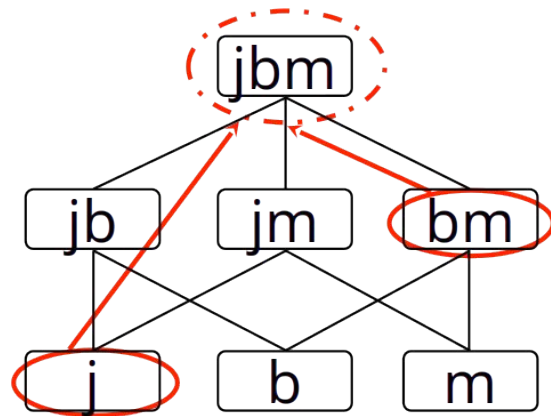
**Closure under
summation**

Domain restrictions: mixed predicates

- Let P_m be the set of mixed predicates (“*carry a piano*”, “*solve the exercise*”, etc.)
 - The domain of P_m is the universe of M : for all $R \in P_m$, $V_M(R) \subseteq U$



Non-distributive



**Closed under
summation**

Language for plural terms

- We extend our logical language with a **summation operator** $\oplus$, a **one-place predicate** **At** for “atom”, and a **two-place relation** $\triangleleft$ for “(proper) individual part”
 - $s \oplus m$ *“the sum consisting of Sally and Mary”*
 - $s \triangleleft s \oplus m$ *“Sally is member of the sum consisting of Sally and Mary”*
 - $s \oplus m \triangleleft c$ *“Sally and Mary are members of the committee”*
- In addition, we introduce:
 - Variables ranging over proper sums: $X, Y, Z, \dots$
 - Number-specific (predicate) constants: $student_{SG}, student_{PL}$

Interpretation of plural terms

- $\llbracket a \oplus b \rrbracket^M = \llbracket a \rrbracket^M \sqcup \llbracket b \rrbracket^M$
- $\llbracket a \triangleleft b \rrbracket^M = 1$ iff $\llbracket a \rrbracket^M \leq \llbracket b \rrbracket^M$ and $\llbracket a \rrbracket^M \neq \llbracket b \rrbracket^M$
- $\llbracket At(a) \rrbracket^M = 1$ iff $\llbracket a \rrbracket^M \in A$
- Individual constants denote either atoms ($x \in A$) or sums ($x \in U - A$)
- Predicate expressions satisfy specific constraints:
 - $V_M(student_{SG}) \subseteq A$
 - $V_M(student_{PL}) \subseteq U - A$

Interpretation of distributive predicates

- Meaning postulate for plural model structure
 - If a distributive predicate P applies to a set $X \subseteq A$, then the full denotation of P is the join semi-lattice **generated by X**
- The denotation of distributive predicates P is uniquely determined by their atomic members:
 - $\forall x[P(x) \leftrightarrow \forall y[At(y) \wedge y \triangleleft x \rightarrow P(y)]]$

Mixed predicates: examples of ambiguous interpretation

- *“every student summarized a paper”*
- *“John and Mary summarized a paper”*
- *“two students summarized a paper”*
- *“John summarized three papers”*

Mass nouns

- **Mass nouns** (“*water*”, “*gold*”, “*wood*”, “*money*”, “*soup*”, etc.) behave like plurals in different respects
- Closure under summation:
 - water + water = water
 - students + students = students
- Combination with cardinalities:
 - “*five liters of water*”
 - “*five bus-loads of students*”
- (Some) shared grammatical patterns:
 - **“*a students are hard workers*”
 - **“*a water is wet*”

Mass nouns vs. plurals

- Unlike plurals, mass nouns are divisive:
 - An amount of water can always be subdivided into proper parts, which are water again
- The denotation of mass nouns cannot be reduced to model theoretic atomic individuals
 - When talking about water, we are not talking about a collection of individual entities

Model structure for mass nouns

- Lets add another sort of entities, the “portions of matter” M , to the model structure, and distinguish a part relation for individuals ($\leq_i$) and a part relation for materials ($\leq_m$):
- $M = ((U, \leq_i), (M, \leq_m), V)$
 - $U \cap M = \emptyset$
 - $(U, \leq_i)$ is an atomic join semi-lattice
 - $(M, \leq_m)$ is a non-atomic and **dense** join semi-lattice
 - V is an interpretation function

Materialization

- There is a close relation between the domain of material entities and the domain of (atomic and sum) individuals: *each individual consists of a specific portion of matter*
- Let $M = ((U, \leq_i), (\mathbb{M}, \leq_m), h, V)$ be a model structure in which $h: (U, \leq_i) \rightarrow (\mathbb{M}, \leq_m)$ is a “materialization” function:
 - h is a **(join semi-)lattice homomorphism** that maps (atomic and plural) individuals to the matter they consist of
 - $a \leq_i b \rightarrow h(a) \leq_m h(b)$
 - $h(a \sqcup_i b) = h(a) \sqcup_m h(b)$

Representation of mass nouns

- Additions to the logical representation language:
 - Variables referring to matter: **x**, **y**, **z**, ...
 - A material fusion operation $\oplus_m$ and a material part relation $\triangleleft_m$ (to be distinguished from $\oplus_i$ and $\triangleleft_i$, respectively)
- A new logical operator m that expresses the materialization function:
 - $\llbracket m(\alpha) \rrbracket^M = h(\llbracket \alpha \rrbracket^M)$, where $\alpha \in WE_e$ is a well-formed expression denoting an individual/group entity—i.e. $\llbracket \alpha \rrbracket^M \in U$

Examples

- “the ring is made of gold”

$$\mapsto \exists y(\text{ring}(y) \wedge \text{gold}(m(y)))$$

- “the ring contains gold”

$$\mapsto \exists x \exists y(\text{ring}(x) \wedge y \triangleleft_m m(x) \wedge \text{gold}(y))$$